



Language: English (README_SUMMARY.md) | Turkce (README_SUMMARY_TR.md)

MertFormer Titan - External Summary (Build 30 V2)

What This Project Is

MertFormer Titan is an offline-first, auditable AI systems repository built around low-bit runtime infrastructure, local assistant foundations, and disciplined evaluation surfaces.

Long-term target: reduce the cost barrier for auditable AI training and inference for compute-constrained developers, small teams, and local institutions. This is a target, not a benchmark claim, until checkpoint-bound runs and target-hardware measurements exist.

Architecture (at a glance)

| Component | What | Why |

|---|---|---|

| BitNet b1.58 | ternary weights (-1 / 0 / +1) | edge memory + energy budget |

| GQA attention | 16 query / 8 KV heads | smaller KV cache, mobile footprint |

| Sparse MoE | 8 experts, top-2 (every 3rd layer) | capacity at ~constant active FLOPs |

| Liquid/CfC mixer | continuous-time recurrence at layers [4, 10, 16] | temporal-dynamics research bet (12-seed ablation: no measured benefit) |

18 layers · hidden 2048 · ~3.67B measured params (2.64B design target). Full detail: ARCHITECTURE.md (ARCHITECTURE.md).

Current Exact State

- Stage: pilot-ready pre-training baseline
- Repo-side readiness: TRAIN_ALLOWED
- Exact reason code: READY_REMOTE_BOOTSTRAP
- Recommended repo-side lane: remote_bootstrap
- Strict local lane: offline_clean
- Preferred serious validation target: 45K
- Application gate: real owned training run + checkpoint-bound evidence
- Remaining non-winning blockers:
offline_clean:PRECOMPUTED_LOGITS_MISSING_AND_PHASE0_NOT_ACTIONABLE,
online_teacher:MISSING_HF_TOKEN

What Matters For Review

- No trained checkpoint claims are made yet.
- Benchmark status remains NOT ELIGIBLE FOR CLAIM until a trained checkpoint exists.
- Exact 45K is preferred, but application readiness is defined by a meaningful real training run with

checkpoint-bound evidence.

- Export/device evidence is a strong plus, not a hard blocker.

Shortest Review Path

1. START_HERE.md (START_HERE.md)
2. docs/PROJECT_MASTER_TRUTH.md (docs/PROJECT_MASTER_TRUTH.md)
3. reports/final_truth_matrix.md (reports/final_truth_matrix.md)
4. reports/known_limits_v1.md (reports/known_limits_v1.md)
5. reports/systems_performance_case_study.md (reports/systems_performance_case_study.md)
6. reports/offline_assistant_case_study.md (reports/offline_assistant_case_study.md)
7. reports/chess_proof_teaching_case_study.md (reports/chess_proof_teaching_case_study.md)
8. STATUS.md (STATUS.md)

Canonical Commands

```
bash scripts/bootstrap_venv.sh
bash scripts/verify_all.sh
bash zero_touch_start.sh --check-only
bash zero_touch_start.sh
bash scripts/final_one_shot.sh
```

Strongest Signals

- real distributed-training debugging — NCCL hang, MoE gradient-checkpoint crash, EOS-masking fixes (ADR-0004) + atomic-checkpoint / resume hardening
- backend routing and fallback honesty
- offline-first, governance-gated assistant foundations
- claim-safe verification and repo truth sync
- compute-accessibility positioning for lower-cost auditable training/inference, with target status kept separate from measured evidence

Open Post-Run Evidence Class

- trained final weights
- best/latest checkpoint proof
- checkpoint-bound benchmark outputs
- trained demo bundle
- trained export/device measurements